Isosceles Right Triangle

An isosceles right triangle is a triangle that has:

- Two equal sides
- One right angle (90°)

It is also called a **45°–45°–90° triangle** because the other two angles are equal.

Properties

- One angle = 90°
- Other two angles = 45° each
- Two sides are equal

Hypotenuse Formula

If each equal side is a, then:

$c = a\sqrt{2}$

where:

- a = equal sides
- c = hypotenuse

Perimeter Formula

$P = 2a + a\sqrt{2}$

or

$P = a(2 + \sqrt{2})$

where:

- a = equal side length

Area Formula

$A = a^2 / 2$

where:

- `A` = area
- `a` = equal side length

Example

If `a = 8 cm`:

- Hypotenuse = `8√2 cm`
- Perimeter = `16 + 8√2 cm`
- Area = `32 cm²`